Requirement Specification:

Enable user GitLab & Docker images

Amit Karpe, Lead Engineer

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Contents

Prepared By

Name	Project Role	Signature	Date
Amit Karpe	Lead Engineer		30 Nov 2024

Reviewed By

Name	Project Role	Signature	Date
Yeo Zhen Xuan	Tech Lead		30 Nov 2024

Approved By

Name	Project Role	Signature	Date
Luke Lim	Synapxe Project Manager		30 Nov 2024
Kevin Lam	TRUST Business Lead		30 Nov 2024

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0.0.1 1. INTRODUCTION

1.1 Document Purpose This document defines the requirement specifications for deploying code, container repositories, and related cloud infrastructure in a non-internet environment within Singapore Government on Commercial Cloud (GCC) by GovTech. The project aims to enable private repository hosting, pipeline execution, and container management for Lifebit CloudOS workflows.

1.2 System Overview Tech Deliverable
4a. Enable Lifebit users to run their own codes

- Phase 1 – Setup repositories (GitLab and AWS ECR) for hosting, accessing & managing private Docker tools & scripts/codes
- Phase 2 – Setup to support hosting, accessing & managing NextFlow pipeline

Problem Statement
Currently, there is no version control repository available within the TRUST-Lifebit platform where researchers can host their custom codes and pipelines.

Outcome
CloudOS user can securely access and perform version control and collaborate with fellow researchers of their private codes, docker tools, and pipelines using Gitlab account and docker images using AWS ECR.

1.3 Project Objectives / Goals

- Enable private code and container repository hosting within a closed cloud environment.
- Provide a scalable and secure solution for pipeline execution and container management.
- Ensure compliance with regulatory and security requirements.

1.4 Definitions/Glossary

- **GitLab Server**: Self-hosted version of GitLab for code repository management.
- **AWS ECR**: Private container image repository for storing and managing Docker images.
- **AWS Batch**: Managed batch processing service for running containerized workloads.
- **CloudOS**: Lifebit's cloud-native bioinformatics platform.

1.5 References

- GovTech GCC Documentation
- AWS Security Best Practices

0.0.2 2. FUNCTIONAL SPECIFICATION OVERVIEW

2.1 In-Scope

- Setup a self-hosted GitLab server within the CloudOS environment.
- Setup of AWS ECR in a separate AWS GCC account for private container storage.
- Integration of AWS Batch for executing workflows.

2.2 Out-of-Scope

- External internet connectivity for GitLab or ECR.
- Support for third-party container registries.

2.3 Assumptions and Dependencies

- AWS GCC accounts are pre-configured with necessary permissions.
- CloudOS is deployed and functional in a closed environment.

2.4 User Access Researchers and engineers will have access to GitLab for version control via Commandline.
AWS ECR will be accessible only within AWS GCC accounts via private networking.

0.0.3 3. WORKFLOWS

Existing or "As-Is" Process Workflow NA

Proposed or "To-Be" Process Workflow NA

0.0.4 4. FUNCTIONAL REQUIREMENTS

4.1 Statutory or Regulatory Requirements NA

4.2 <Functional Requirement or Feature # 1> NA

4.3 <Functional Requirement or Feature # 2> NA

0.0.5 5. DATA SETUP

NA

0.0.6 6. RECONCILIATION REPORTS

NA

0.0.7 7. BATCH JOBS

NA

0.0.8 8. AUDIT LOGGING

NA

0.0.9 9. DATA MIGRATION REQUIREMENTS

NA

0.0.10 10. INTERFACE REQUIREMENTS

NA

0.0.11 11. PERFORMANCE REQUIREMENTS

NA

0.0.12 12. SUMMARY / RECOMMENDATIONS

NA
